

hexis_ctree — Rissanen-style MDL context tree for HEXIS

Reference implementation of the single instrument of the HEXIS project (Decision D32): a variable-order context-tree model following **Schürmann & Grassberger (1996)**, *Entropy estimation of symbol sequences*, Chaos 6(3):414–427 (arXiv:cond-mat/0203436), hereafter **S&G**, aligned with the frozen HEXIS Master Specification §4.1–§4.4 (primary configuration C0, Decision D18) and extended with the HEXIS-specific requirements (sentence-level hard reset, frozen out-of-sample evaluation, label-free P2/S1 score construction, position restriction `available_past ≥ 4`).

Pure Python ≥ 3.9 , standard library only. Deterministic. No network, no third-party dependencies. Status: **engine validated on synthetic sources; Phase 0 (corpus acquisition) not started; no claim in this package concerns the Perseus treebanks**. Every implementation-level reading of the spec and every divergence between C0 and the S&G-published choices is itemized in `ASSUMPTIONS.md` (Decision-Log ratification material, feeds Task T5.1).

Configuration: C0 default, SG96 preset

`ContextTreeConfig()` defaults to the frozen HEXIS primary configuration C0 (Spec §4.1/§6.3, D18): `beta = 0.5`, `k_min = 2`, `delta_threshold (= γ) = 0.0`, `max_depth (= d_max) = 8`, `growth = "eager"` (the spec's training pass grows a node on first visit), monotone-stop selection. Named constructors:

- `ContextTreeConfig.hexis_c0(max_depth=8)` — explicit C0 (use `max_depth=6` for the Latin arm, D24);
- `ContextTreeConfig.sg96_published(max_depth=8)` — the choices published by S&G, for the Task T5.1 correspondence audit: `beta = 1/d` (Eq. 30), Rissanen frugal growth (Sec. IV), no count floor (`k_min = 1`).

Verified correspondence with S&G

Every algorithmic component below was checked against the arXiv full text of S&G before implementation (fetch date: 2026-07-07).

Component	S&G locus	HEXIS locus	Code locus
Conditional estimator $\hat{p}(a \mid s) = (n_s(a) + \beta) / (N_s + \beta d)$	Eq. (21)	Spec §4.1 (identical)	<code>model._log2_prob</code>
β value	Eq. (30): $\beta = 1/d$, <i>empirical</i> finding on written English	D18: $\beta = 0.5$ (KT- style); sensitivity axis D37	<code>config</code> (C0 default 0.5; SG96 preset 1/d)
Prequential code-length difference per node, updated recursively	Sec. V.B, Eqs. (22)– (24)	Spec §4.1: $\Delta(s) = L_{\text{par}} - L_{\text{self}}$; stored here with the	<code>model._Delta_update</code>

Component	S&G locus	HEXIS locus	Code locus
with pre-update counts		S&G sign ($\Delta = L_{\text{self}} - L_{\text{par}}$), so $\Delta_{\text{spec}} = -\Delta$ — conditions exactly equivalent	
Context selection with threshold	Sec. V.B: shortest context with $\Delta < -\delta$, $\delta \geq 0$ application-dependent	Spec §4.1 monotone-stop: deepest chain with $\Delta_{\text{spec}} > \gamma$ and $\text{total} \geq k_{\text{min}}$; $\gamma = 0$, $k_{\text{min}} = 2$ (D18). k_{min} has no S&G counterpart ($k_{\text{min}} = 1$ disables it)	<code>model._select_index</code>
Tree growth	Sec. IV (end): create one node when the deepest-matched count of the current symbol reaches 2 ("rissanen")	Spec §4.1 pseudocode: <code>setdefault</code> on first visit ("eager", C0 default)	<code>model.fit</code> , step (5)
Depth cap	Sec. VI.B sanctions truncated trees (upper bounds)	D18: $d_{\text{max}} = 8$ (Latin 6, D24); sensitivity axis D37	<code>config.max_depth</code> <code>h</code>
Online estimator $\hat{h} = (1/N) \sum -\log_2 \hat{p}(s_t \text{selected})$	Secs. V.B, V.D	Spec §4.5 h_{online} (descriptive only, D19/D32)	<code>model.fit</code> step (2)
Finite-N convergence $\hat{h}_N \approx h + c \cdot \log N/N^\gamma$ (slow, from above; empirical ansatz)	Eq. (28)	motivates the matched-N, relative-only design (D19/D20)	documented caveat

Package layout

```

hexis_ctree/
  config.py      ContextTreeConfig: C0 defaults; hexis_c0 / sg96_published
presets
  alphabet.py    Alphabet: fixed ex ante, deterministic, fail-loud
  model.py       ContextTreeModel: online fit (S&G) + frozen scoring (HEXIS)
  scores.py      Per-position records (codelen, root codelen, depth_selected,
                 depth_matched, available_past); the  $\geq 4$  mask; aggregators
  stats.py       Reference statistic layer: P1 transfer/LOD0; label-free P2/S1
                 (spec §4.4 unweighted document means); document-level
                 permutation test (Monte Carlo stand-in for the exact schemes)
  synthetic.py   Seeded generators with analytical ground truth (validation)
tests/
examples/
  demo_synthetic.py Full P1/P2/S1 pipeline on a known-truth toy corpus
run_tests.py     Dependency-free runner (use `pytest tests/` when available)

```

Quickstart

```
from hexis_ctree import (Alphabet, ContextTreeConfig, ContextTreeModel,
                        mean_context_gain_bits, mean_selected_depth)

# documents = list of sentences; sentence = list of UPOS+DEPREL symbol strings
# (post-normalization; empty sentences and out-of-alphabet symbols are errors)
alphabet = Alphabet.from_documents(all_documents) # fixed ex ante, pooled
model = ContextTreeModel(alphabet) # config defaults to C0
model.fit(training_documents) # single online pass (S&G)
h_online = model.online_entropy_rate # descriptive  $\hat{h}$ , bits/token
model.freeze()

scores = model.score_document(some_document) # frozen per-position records
ce = model.cross_entropy(eval_documents) # frozen mean code length (P1)
gain = mean_context_gain_bits(scores) # P2 ingredient (past  $\geq 4$ )
depth = mean_selected_depth(scores) # S1 ingredient (past  $\geq 4$ )
```

Statistic layer (reference implementation; protocol orchestration — T*-matched subsampling, seeds, exact enumeration — belongs to the protocols phase):

```
from hexis_ctree import (fit_context_tree, transfer_matrix,
                        pooled_document_scores, per_document_gain,
                        permutation_test_label_free)

tm = transfer_matrix({"HEX": hex_docs, "PROSE": prose_docs},
                    alphabet, lodo_diagonal=True) # P1 display
pooled = fit_context_tree(all_docs_list, alphabet) # label-free
gains = per_document_gain(pooled_document_scores(pooled, docs_by_id))
t, p = permutation_test_label_free(gains, labels, "HEX", "PROSE",
                                   n_permutations=10_000, seed=0)
# default statistic = spec §4.4 form (unweighted mean over documents)
```

Semantics guaranteed by the implementation (and by tests)

1. **Sentence-level hard reset (D10 P-RESET).** The context never crosses a sentence boundary; no BOS symbol exists anywhere; all averages run over real tokens only; `available_past` at position i equals i .
2. **Position restriction (D35).** P2 (gain) and S1 (depth) aggregate only positions with `available_past  $\geq 4$` ; P1 uses all real tokens (position effects cancel within-document, per spec §4.4).
3. **Monotone-stop selection (D18).** A node is trusted only if every node on its chain saves bits ($\Delta_{\text{spec}} > \gamma$) and has `total  $\geq k_{\text{min}}$` ; applied identically online (pre-update counts) and frozen.
4. **Frozen evaluation (§4.3).** `score_*` and `cross_entropy` never mutate the model (tested byte-for-byte); per-position records carry (`codelen_selected`, `codelen_root`, `depth_selected`, `depth_matched`, `available_past`); ancestor fallback is implicit in the monotone descent (root always exists); in-alphabet unseen symbols get β -smoothed positive probability; out-of-alphabet symbols raise (alphabet is frozen at G1).
5. **Label-free P2/S1 (D36).** The pooled model and all per-position scores are computed once without labels; permutations reassigned labels over fixed per-document values.

6. **Determinism (D26-compatible).** No randomness in fitting; alphabet ordering is a pure function of the symbol set; generators and the permutation test are seeded.
7. **Units.** All code lengths and rates in bits (log base 2), per spec §0.3.

Validation results (synthetic sources, seeded, reproducible)

Both the frozen C0 configuration and the SG96-published preset were run on all three sources; on these clean synthetic processes the two are nearly indistinguishable — evidence that the C0-vs-S&G divergences (β , growth policy, $k_{\min}$) do not distort the instrument where the truth is known.

Source	Ground truth	Config	Online $\hat{h}$	Frozen CE (fresh)	Mean gain	Mean depth
IID uniform, $d=4$, $N=50k$	$h = 2.0000$	C0	2.0006	2.0000	+0.0000	0.000
		SG96	2.0006	2.0000	+0.0000	0.000
Markov order-1, $d=2$, $p_{\text{stay}}=0.9$, $N=60k$	$h = 0.4690$; $H_1 - h = 0.5310$	C0	0.4754	0.4726	+0.5289	1.000
		SG96	0.4754	0.4726	+0.5289	1.000
Noisy template abacad, $d=6$, $\epsilon=0.02$, $N=24k$	phase-known $h = 0.1879$	C0	0.2690	0.3130	+1.5659	1.997
		SG96	0.2675	0.3122	+1.5668	1.998

The IID case is the key negative control: tens of thousands of candidate nodes are grown, yet the MDL gate selects depth 0 everywhere and the measured gain is exactly zero. The Markov case locks depth at exactly 1 with gain within 0.003 bits of the analytical value. In the end-to-end demo the PROSE LODO diagonal (2.2740 bits) reproduces the analytical entropy rate (2.2698 bits) to +0.004 bits. Excesses over ground truth are consistent with S&G Eq. (28) plus the per-sentence restart cost.

Run everything with:

```
python3 run_tests.py # or: pytest tests/
python3 examples/demo_synthetic.py
```

Caveats to carry into the manuscript

- **Eq. (28) convergence.** The online estimate exceeds h at finite N and decays slowly ($c \cdot \log N / N^\gamma$, empirical ansatz in S&G). HEXIS therefore compares regimes **relatively at matched N** (T^* -matching, D20) and never reports absolute entropy rates.
- **$\beta = 1/d$ is an empirical recommendation in S&G** (numerically superior on English text), not a theoretical optimum; C0 fixes $\beta = 0.5$ and β is a declared sensitivity axis (D37).

- **Pooled-protocol interpretation.** Under the label-free pooled model, the gain baseline is the pooled order-0 marginal and selected depths are relative to the pooled mixture source; deep contexts may partly encode *regime identity* in addition to regime-internal memory. This does not affect the validity of the permutation test (fully label-free) but must inform the interpretation of P2/S1 magnitudes.
- **Heterogeneity.** S&G document (their Bible example) that statistically inhomogeneous corpora inflate online estimates; this supports the HEXIS document-level LODO design.

Explicitly out of scope (by design)

CoNLL-U parsing and alphabet normalization (pipeline Phase 1); protocol orchestration (gates G0–G7, protocols (a)/(b)/(c) with T*-matched whole-sentence subsampling and $S = 20$ seeds, the 13-cell sensitivity plan of D37); the `select = argmax` sensitivity variant (one D37 cell — pending); exact enumeration tests (sign-flip 2^{11} , label permutation $C(11,5)$ — the Monte Carlo shuffle here is an engine-validation stand-in); JSD/R1; hierarchical bootstrap; the context lexicon (D38); any corpus claim.